

Leon Mayer

Leon.mayer@dkfz-heidelberg.de
leonmayer.github.io
www.linkedin.com/in/leond-mayer
scholar.google.com/citations?user=uYVPJyIAAAAJ

EDUCATION

PhD Student

Since October 2023

Heidelberg University

- Topic: Vision-Language Foundation Models for Surgery
- Created a surgical VLM benchmark dataset and conducted a broad evaluation study across state-of-the-art models, revealing critical failure modes
- Currently focusing on LLM explainability for medicine

Doctoral Visit

Nov 2025-Dec 2025

Hong Kong University of Science and Technology

- Co-developed a large-scale self-supervised video foundation model for intelligent surgery

Electrical Engineering and Information Technology, M.Sc.

2020-2023

Technical University of Munich

- Area of specialization: Bioengineering and Neuroengineering
- Final Grade: 1.4
- Spent an exchange semester at the Danish Technical University in Copenhagen working on predicting postoperative complications
- Thesis on utilizing Deep Learning for anticipating instrument changes during laparoscopic surgeries

Electrical Engineering, Information Technology and Computer Engineering, B.Sc.

October 2016 – April 2020

RWTH Aachen University

- Area of specialization: Micro- and Nanoelectronics
- Thesis on patient temperature measurement in infrared ICU images

WORK EXPERIENCE & PROJECTS

Doctoral Researcher

Since October 2023

Div. Intelligent Medical Systems, German Cancer Research Center

- Supervised multiple students, including Bachelor's and Master's theses
- Participating in interdisciplinary courses on the biology of cancer
- Part of winning team at MICCAI 2024 (Critical View of Safety)

Erasmus semester project

September 2021 – January 2022

Technical University of Denmark, Copenhagen

- Developed machine learning models to predict serious adverse events from post-surgical patient vital signs
- Collaborated in an international team of engineers and clinicians
- First-author paper presented at IEEE EMBC 2022

Computer Vision internship

June 2020 – September 2020

Audi AG, Ingolstadt

- Created an image processing application for automated front lighting evaluation using Python and JavaScript

Research assistant

November 2018 – August 2019

Chair for Medical Signal Processing, RWTH Aachen

- Implemented a sensor-based heartbeat detection system embedded in a seat cushion
- Designed and manufactured PCB boards for the prototype

SKILLS & INTERESTS

- **Languages:** German – Native Proficiency
English – C1, Cambridge CAE
- **Programming languages:** Python, C/C++, JavaScript
- **Courses:** BlueDot Impact AGI Strategy Course
- **Tools:** Huggingface Transformers, Pytorch, Git, Docker, HPC, CVAT, Claude Code
- **Interests:** Table tennis, robotics, reading

PUBLICATIONS

- **Mayer, L.,** Kalinowski, P., Ebersbach, C., Knopp, M., Rädtsch, T., Christodoulou, E., ... & Maier-Hein, L. (2025). 6 Fingers, 1 Kidney: Natural Adversarial Medical Images Reveal Critical Weaknesses of Vision-Language Models. *arXiv preprint* arXiv:2512.04238.
- Kröger, J., Jorek, N., Seitel, A., **Mayer, L.,** Salg, G. A., Crnovrsanin, N., ... & Nienhüser, H. (2025). Application of artificial intelligence in esophageal surgery: a systematic review. *Journal of robotic surgery*, 19(1), 694.

- **Mayer, L.**, Rädtsch, T., Michael, D., Luttner, L., Yamlahi, A., Christodoulou, E., ... & Maier-Hein, L. (2025). Challenging Vision-Language Models with Surgical Data: A New Dataset and Broad Benchmarking Study. *arXiv preprint* arXiv:2506.06232.
- Yang, S., Zhou, F., **Mayer, L.**, Huang, F., Chen, Y., Wang, Y., ... & Chen, H. (2026). Large-scale self-supervised video foundation model for intelligent surgery. *npj Digital Medicine*.
- Rädtsch, T., **Mayer, L.**, Pavicic, S., Kavur, A. E., Knopp, M., Öztürk, B., ... & Maier-Hein, L. (2025). Bridging vision language model (VLM) evaluation gaps with a framework for scalable and cost-effective benchmark generation. *arXiv preprint* arXiv:2502.15563.
- Mølgaard, J., **Mayer, L.**, Rasmussen, S. S., Haahr-Raunkjær, C., Achiam, M. P., Jørgensen, L. N., ... & Aasvang, E. K. (2025). Nadir heart rate as a novel risk marker of postoperative complications-A propensity matched analysis. *Sleep Medicine*, 128, 65-73.
- Christodoulou, E., Reinke, A., Houhou, R., Kalinowski, P., Erkan, S., ..., **Mayer, L.**, ... & Maier-Hein, L. (2024, October). Confidence intervals uncovered: Are we ready for real-world medical imaging AI?. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 124-132). Cham: Springer Nature Switzerland.
- Wagner, L., Schneider, D. N., **Mayer, L.**, Jell, A., Müller, C., Lenz, A., ... & Wilhelm, D. (2024). Towards multimodal graph neural networks for surgical instrument anticipation. *International journal of computer assisted radiology and surgery*, 19(10), 1929-1937.
- Wagner, L., Jourdan, S., **Mayer, L.**, Müller, C., Bernhard, L., Kolb, S., ... & Wilhelm, D. (2024). Robotic scrub nurse to anticipate surgical instruments based on real-time laparoscopic video analysis. *Communications Medicine*, 4(1), 156.
- **Mayer, L.**, Gu, Y., Aasvang, E. K., & Meyhoff, C. S. (2022, July). Prediction of Serious Adverse Events from Nighttime Vital Signs Values. *44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)* (pp. 2631-2634). IEEE.
- Lyra, S., **Mayer, L.**, Ou, L., Chen, D., Timms, P., Tay, A., ... & Hoog Antink, C. (2021). A deep learning-based camera approach for vital sign monitoring using thermography images for ICU patients. *Sensors*, 21(4), 1495.
- Loh, K., Hoog Antink, C., Bernhard, C., **Mayer, L.** & Feld, J. (2020). Child-appropriate experiment on auditory selective attention in a virtual acoustic environment. *Proceedings of the Fortschritte der Akustik: DAGA*